



PART A: General Information

1. Course Number **CE 206**
Course Title **Engineering Computation Sessional**
Credit (Contact) Hours 1.5 (3.0)
2. Level and Term (Section) Level-2, Term-2 (C2)
Academic Term July 2021
3. Type of Course Basic Engineering
Offered to Civil Engineering
4. Pre-requisite Course(s) CE 204
5. Course Website <https://teams.microsoft.com>
6. Lecture Schedule Saturday (2:30-5:00 PM) – Online (Zoom meeting platform)
Monday (11:00 AM-1:30 PM) –On-campus (CE-304;UG Computer Lab)
7. Important Dates For important dates and examination schedules and latest updates, please follow the course website
8. Course Teacher(s)

Name (Initials):	Office:	Email:	Consultation Hour(s)
Dr. Sheikh Mokhlesur Rahman	JRC-CE 503	smrahman@ce.buet.ac.bd	Monday (3:00-4:00 PM)
Tazwar Bakhtiyar Zahid	JRC-CE 643	tazwarzahid@ce.buet.ac.bd	--

PART B: Course Details

9. Course Content (As approved by the Academic Council)
Introduction to hi-level computational programming tools; application to numerical analysis: basic matrix computation, solving systems of linear equations, non-linear equations, differential equations, interpolation and curve fitting, numerical differentiation, numerical integration; application to engineering problems: solving problems related to mechanics, numerical solution of equation of motion etc.
10. Course Objectives
 - To teach the students fundamental of MATLAB programming and its application in engineering field
 - To prepare the students so that they can apply MATLAB in their problem analysis and solving.
11. Knowledge required
Basic programming knowledge in any language, linear algebra
12. Course Outcomes
At the end of the course a student should be able to

CO No.	CO Statement	Corresponding PO(s)*	Domains and Taxonomy level(s)**	Delivery Method(s) and Activity(-ies)	Assessment Tool(s)
1	write basic codes to solve programming problems using MATLAB	PO(e)	C1	Lectures, Homework	hands-on programming; examination; assignment
2	demonstrate competence in implementing hi-level programming tools to solve numerical computational problems (CO2)	PO(e)	C3, A3, P5	Lectures, Homework	hands-on programming quiz, assignment

3	apply hi-level computational tools to solve civil engineering related problems (CO3)	PO(b) PO(e)	C3	Lectures, Homework	hands-on programming, project
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*PO(a): Engineering knowledge; PO(b): Problem analysis; PO(c): Design/development of solutions; PO(d): Investigation; PO(e) Modern tool use; PO(f): The engineer and society; PO(g): Environment and sustainability; PO(h): Ethics; PO(i): Individual work and teamwork; PO(j): Communication; PO(k): Project management and finance; PO(l): life-long learning

**C-Cognitive: C1: Knowledge; C2: Comprehension; C3: Application; C4: Analysis; C5: Synthesis; C6: Evaluation

A-Affective: A1: Receiving; A2: Responding; A3: Valuing; A4: Organizing; A5: Characterizing

P-Psychomotor: P1: Perception; P2: Set; P3: Guided Response; P4: Mechanism; P5: Complex Overt Response; P6:

Adaptation; P7: Organization

13. Lecture Plan

Class No.	Topics	References	Corresponding CO(s)
Lecture 1	MATLAB Fundamentals: Basic MATLAB syntax and commands, arrays, vectors, and matrix operation, plotting	Handout	CO1
Lecture 2	MATLAB Fundamentals: Loops- <i>while</i> , <i>for</i> , <i>if-else</i> scripting, function, file input-output <i>Class Assessment 1</i>	Handout	CO1
Lecture 3	Roots of equations <i>Group formation for Project (4 Students/group)</i>	Handout Chapra (Chapter 5 & 6)	CO1 CO2
Lecture 4	Roots of equations <i>Class Assessment 2</i>	Handout Chapra (Chapter 5 & 6)	CO1 CO2
Lecture 5	System of linear equations	Handout Chapra (Chapter 9 & 10)	CO1 CO2
Lecture 6	Curve fitting and interpolation <i>Project proposal submission</i> <i>Class Assessment 3</i>	Handout Chapra (Chapter 14 & 17)	CO1 CO2 CO3
Lecture 7	Numerical differentiation and integration	Chapra (Chapter 19, 20, & 21)	CO1 CO2
Lecture 8	Solving ODE <i>Class Assessment 4</i>	Handout Chapra (Chapter 22, 23, & 24)	CO1 CO2
Lecture 9	Fourier Transform; Eigenvalues and eigenvectors	Handout Chapra (Chapter 13, & 16)	CO1 CO2
Lecture 10	Basic statistics: descriptive statistics, distribution, hypothesis testing, regression model, ANOVA <i>Class Assessment 5</i>	Handout	CO1 CO2
Lecture 11	Project presentation (8-10 minutes) + submission	--	CO3
Lecture 12 Week 10 or 11	Quiz 1 (Hands-on Programming)	--	CO2 CO3
Week 13	Quiz 2 (Written Test)	--	CO1 CO2

14. Assessment Strategy

- Class Participation: Class participation and attendance will be recorded in every class.
- Continuous Assessment:
 - 4 to 5 hands-on Class Assessments using MATLAB
 - Assessment will be based on either the topic taught on that day or topic taught on the previous day or combination of the two topics
 - In addition to hands-on assessment, MCQ type assessments may be conducted in each class using google form/ google classroom.
 - Assignment and/or homework will be provided to ensure the active engagement even outside class hours.
- Quizzes: Two nos quizzes will be administered. One of them will be a hands-on demonstration of coding and another one will be written exam.
- Project: A project will be assigned. The project should cover fundamental numerical analysis as well as their relevant application in the Civil Engineering field.

15. Distribution of Marks

Item	Marks
Attendance	10%
Assignment/ Homework/ File	10%
Class Assessment	30%
Quizzes (2 Nos: Hands-on and Written)	35%
Project	15%

16. Textbook

1. **Applied Numerical Methods with MATLAB for Engineers and Scientists**, 4th edition, by Steven C. Chapra, McGraw-Hill Education. (Chapra)
2. Class Lecture Handout

17. References

1. **MATLAB Primer**, Mathworks Inc.

Important University Policies:

- Rules and regulations for the undergraduate programmes:
<https://www.buet.ac.bd/info/Academicinformation/RulesUndergradprogram>

Important Departmental/Course Policies

- Program Outcomes: Link to program outcomes
- Learning Domains: Link to learning domains document

Prepared by:	
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Signature:	Signature:
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Date of Approval by BUGS: NA	